EXPERIMENT NO 7

AIM: Simulation of RC phase-shift oscillator using Pspice.

PROBLEM STATEMENT: To design and analyse a transistor-based RC phase-shift oscillator using PSpice and determine its frequency of oscillation.

SOFTWARE REQUIRED: PSpice Simulator

COMPONENT REQUIRED:

- NPN transistor, e.g. 2N2222 or BC547
- Resistors
- Capacitors
- DC power supply
- Ground

THEORY:

A phase-shift oscillator is an electronic oscillator that generates a sinusoidal signal using an amplifying transistor and an RC feedback network.The transistor provides a phase shift of approximately 180° between its input and output. The three RC sections in the feedback network provide another 180° phase shift at the oscillation frequency.

Therefore, the total phase shift around the feedback loop is: 180° + 180° =360°

or equivalently , satisfying the Barkhausen phase condition.

For a three-section RC phase-shift oscillator with equal values of and , the approximate frequency of oscillation is:

$$f_o = \frac{1}{2\pi RC\sqrt{6}}$$

The amplifier must provide sufficient gain to compensate for the attenuation of the RC feedback network. The minimum theoretical gain is approximately:

$$A_v \geq 29$$

for the ideal equal-component three-section network.

CIRCUIT DIAGRAM:

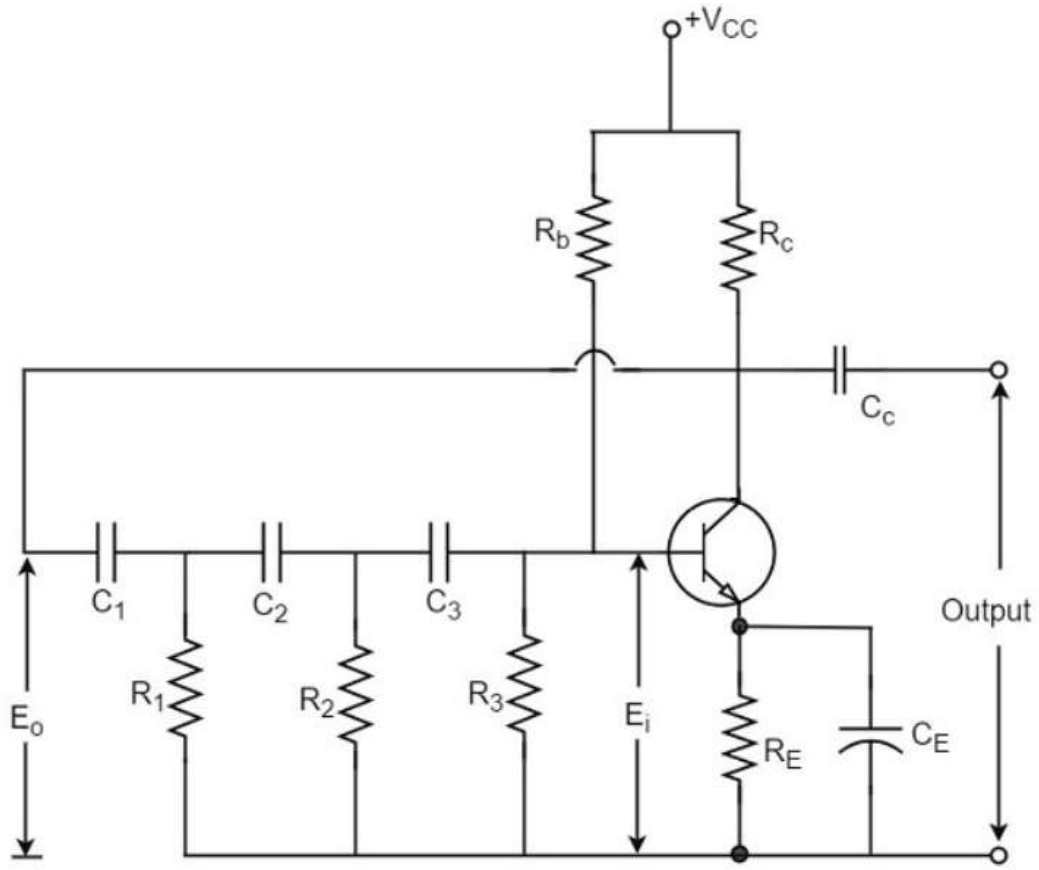


Fig: RC Phase shift Oscillator

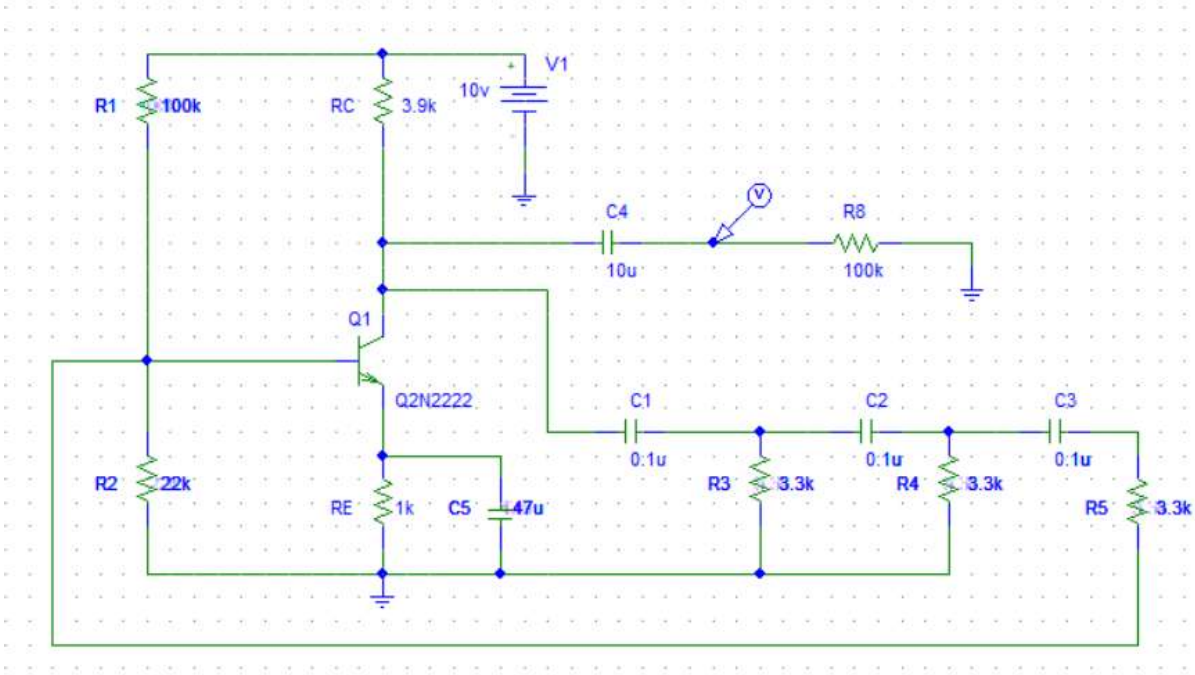


Fig: RC Phase shift Oscillator using pspice

PROCEDURE:

1. Create a New Project

   Select File → New → Project.

2. Place Components as per the circuit diagram.

3.Connect the transistor in common-emitter configuration.

4. Construct the RC Feedback Network

5. Apply the required DC supply

6. Set Transient Analysis

- Print Step: 10u
- Final Time: 400m
- No Print Delay: 0
- Step Ceiling: 10u

7. Run the Simulation and observe the collector voltage using the PSpice Probe.

OBSERVATIONS:

Theoretical Calculation:

R=10k , C=0.1u

$$f_o = \frac{1}{2\pi RC\sqrt{6}}$$

Practical Calculation:

From the simulated waveform, measure the time period T between two successive peaks.

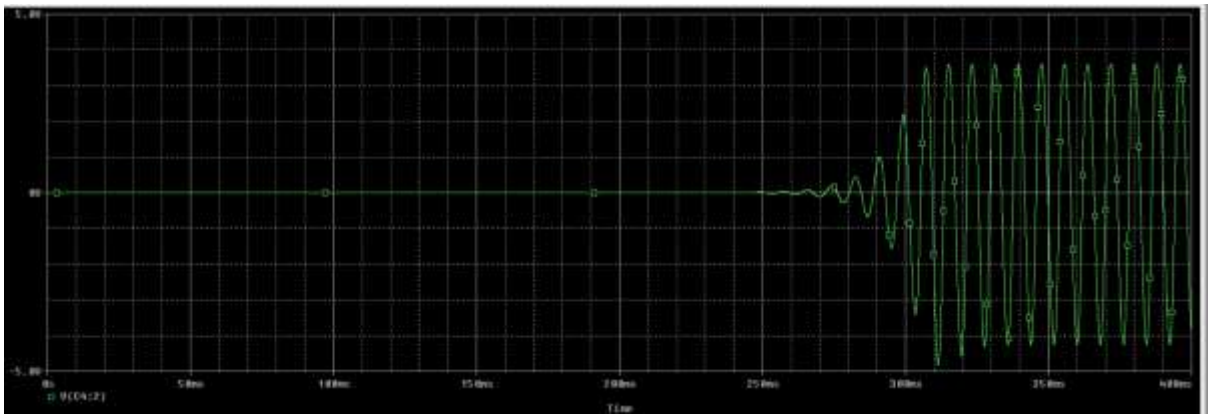
Then calculate:

$$f_o = \frac{1}{T}$$

Percentage Error

$$\%\ Error = \frac{|f_{theoretical} - f_{simulated}|}{f_{theoretical}} \times 100$$

OUTPUT:



RESULT

The transistor-based RC phase-shift oscillator was successfully designed and simulated using PSpice. A sinusoidal output waveform was obtained, and the simulated frequency of oscillation was determined and compared with the theoretical value.

VIVA VOCE:

1. What is an oscillator?

2. What is a phase-shift oscillator?

3. Why is a transistor used in this oscillator?

4. Why are three RC sections used?

5. What is the Barkhausen criterion?